

UNIT – 2
ASSESSMENT – 3
OUTPUT

1.

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A* SEARCH ALGORITHM

1. Use Faculty Graph
2. Enter Custom Graph

Enter Choice : 5

Enter Number of Nodes : 4
Node Name : A
Node Name : B
Node Name : C
Node Name : D

Enter Number of Edges : 6

Edge 1
From : A
To : D
Cost : 5

Edge 2
From : A
To : C
Cost : 3

Edge 3
From : D
To : A
Cost : 6

Edge 4
From : D
To : B
Cost : 5
```

2.

Beta = 6
Current Minimum = 6

Leaf Node : 4
Alpha = 2
Beta = 4
Current Minimum = 4

RIGHT MIN returns : 4

Iteration 3
Evaluating ROOT MAX

Left Value : 2
Right Value : 4

MAX chooses : 4

===== FINAL RESULT =====
Best Move : RIGHT Subtree
Final Minimax Value : 4
Pruned Nodes : None

===== MINIMAX WITH ALPHA-BETA PRUNING =====

Iteration 1
Evaluating LEFT MIN Node

Leaf Node : 5
Alpha = -inf
Beta = 5
Current Minimum = 5

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Leaf Node : 3
Alpha = -inf
Beta = 3
Current Minimum = 3

Leaf Node : 2
Alpha = -inf
Beta = 2
Current Minimum = 2

LEFT MIN returns : 2

MAX updates Alpha = 2

Iteration 2
Evaluating RIGHT MIN Node

Leaf Node : 9
Alpha = 2
Beta = 9
Current Minimum = 9

Leaf Node : 6
Alpha = 2

Beta = 6
Current Minimum = 6

Leaf Node : 4
Alpha = 2
Beta = 4
Current Minimum = 4

RIGHT MIN returns : 4

Iteration 3
Evaluating ROOT MAX

Left Value : 2
Right Value : 4

MAX chooses : 4

===== FINAL RESULT =====
Best Move : RIGHT Subtree
Final Minimax Value : 4
Pruned Nodes : None